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Date: July 30, 2026

Editor-in-Chief
International Journal of Disaster Risk Reduction (IJDRR)
Elsevier B.V.

Subject: Manuscript Submission — "AEGIS: Evidential Multi-Agent Orchestration for Multimodal Urban Flood Risk Assessment: A Subjective Logic Framework with Provenance-Aware Explanations"

Dear Editor-in-Chief,

We are pleased to submit our original research manuscript titled **"AEGIS: Evidential Multi-Agent Orchestration for Multimodal Urban Flood Risk Assessment: A Subjective Logic Framework with Provenance-Aware Explanations"** for consideration as a regular research article in *International Journal of Disaster Risk Reduction* (IJDRR).

Operational flood decision-support systems must fuse heterogeneous multi-modal data streams (satellite radar, optical imagery, reanalysis climate forecasts, terrain rasters, gauge networks, and unstructured bulletins) under frequent sensor dropout and inter-source disagreement. Contemporary LLM-arbitrated agentic pipelines suffer from four critical operational vulnerabilities: (1) minority-class collapse on critical inundation states, (2) uncalibrated probability estimation and silent hallucination under missing modalities, (3) lack of per-agent credibility propagation, and (4) absence of source-level provenance trails.

In this work, we present **AEGIS** (Agentic Evidential Geographic Intelligence for Sustainability), a novel framework interfacing Subjective Logic operator algebra with a specialist five-agent catalog over a shared four-state Dirichlet frame. The key contributions of our study include:

- **Evidential Multi-Agent Orchestration**: Combining Consensus & Compromise Fusion (CCF) and Weighted Belief Fusion (WBF) driven by a dynamic Jensen–Shannon divergence signal.
- **Monotonic Uncertainty Guarantee**: Utilizing a partial-observable projection rule that mathematically guarantees epistemic uncertainty u grows monotonically as modalities drop.
- **Brier-Score Credibility Propagation**: Online tracking of agent reputation to weight fusion decisions dynamically without manual threshold tuning.
- **Provenance-Aware Explainability**: A DuckDB-backed provenance ledger powering an interactive spatial dashboard for source-level auditing.

We evaluate AEGIS on the benchmark Brisbane February–March 2022 flood fold (200 cells $\times$ 24 days $= 4,800$ cell-day instances). AEGIS-SL achieves an F1-Macro of 0.7190, AUROC of 0.9310, and Expected Calibration Error (ECE) of 0.043, outperforming monolithic raw-feature late-fusion baselines and demonstrating strong robustness against minority-class collapse where LLM arbiters fail.

We confirm that:

1. This manuscript is original, has not been published previously, and is not currently under consideration for publication elsewhere.
2. All co-authors (Ali Hamza and Ghulam Mujtaba) have read, approved, and agreed to the submission of this manuscript to *International Journal of Disaster Risk Reduction*.
3. There are no financial or personal conflicts of interest to declare.

Thank you for considering our work. We look forward to hearing from you regarding the review process.

Sincerely,

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